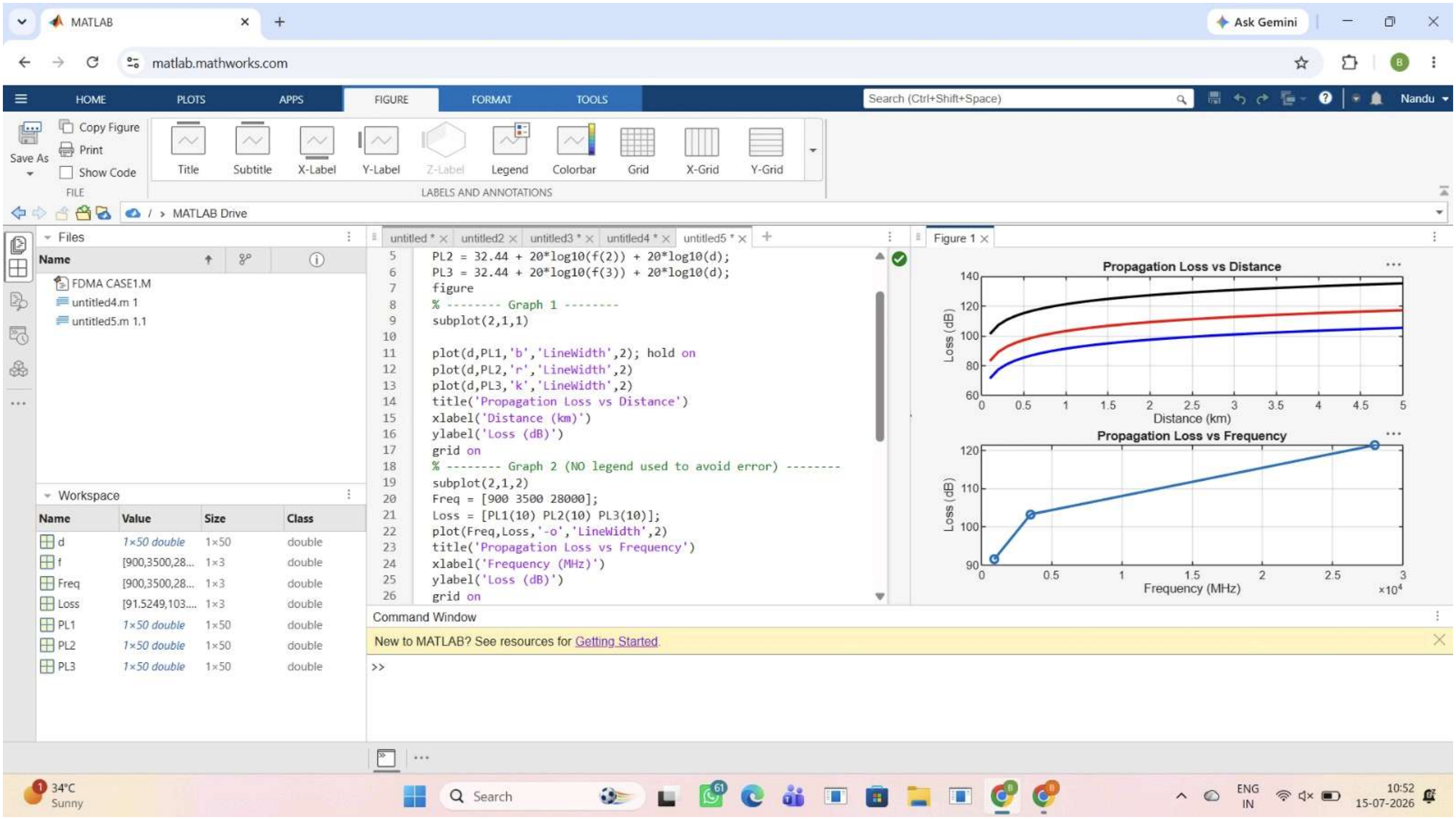

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LABELS AND ANNOTATIONS
MATLAB Drive
Files
FDMA CASE1.M
untitled4.m 1
untitled5.m 1.1
Workspace
Name Value Size Class
d 1×50 double 1×50 double
f [900,3500,28... 1×3 double
Freq [900,3500,28... 1×3 double
Loss [91.5249,103.... 1×3 double
PL1 1×50 double 1×50 double
PL2 1×50 double 1×50 double
PL3 1×50 double 1×50 double
untitled * untitled2 untitled3 * untitled4 * untitled5 *
PL2 = 32.44 + 20*log10(f(2)) + 20*log10(d);
PL3 = 32.44 + 20*log10(f(3)) + 20*log10(d);
figure
% -------- Graph 1 --------
subplot(2,1,1)
plot(d,PL1,'b','LineWidth',2); hold on
plot(d,PL2,'r','LineWidth',2)
plot(d,PL3,'k','LineWidth',2)
title('Propagation Loss vs Distance')
xlabel('Distance (km)')
ylabel('Loss (dB)')
grid on
% -------- Graph 2 (NO legend used to avoid error) --------
subplot(2,1,2)
Freq = [900 3500 28000];
Loss = [PL1(10) PL2(10) PL3(10)];
plot(Freq,Loss,'-o','LineWidth',2)
title('Propagation Loss vs Frequency')
xlabel('Frequency (MHz)')
ylabel('Loss (dB)')
grid on
Figure 1
Propagation Loss vs Distance
Loss (dB)
Distance (km)
Propagation Loss vs Frequency
Loss (dB)
Frequency (MHz)
×10^4
Command Window
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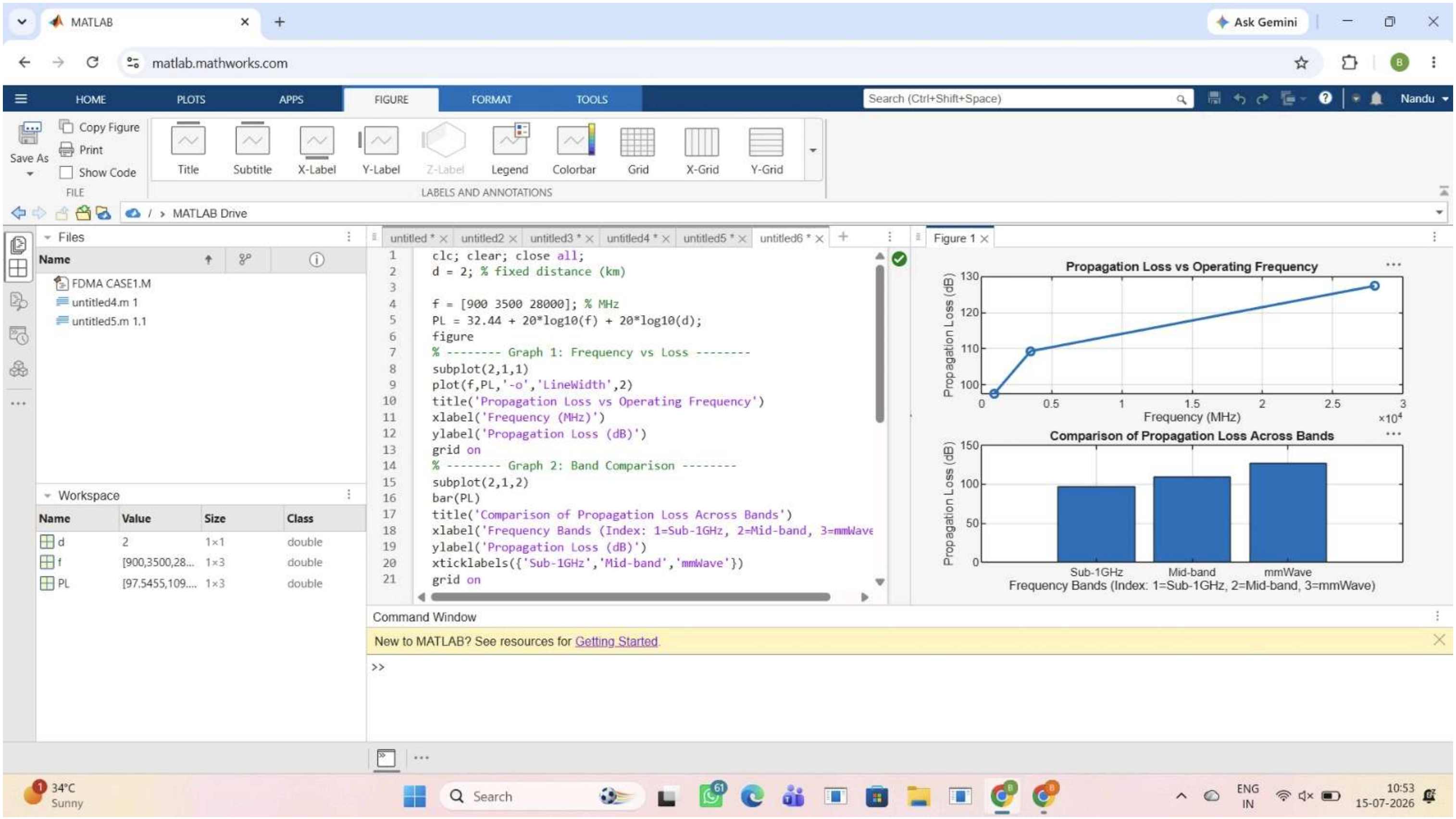
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FILE
LABELS AND ANNOTATIONS
MATLAB Drive
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FDMA CASE1.M
untitled4.m 1
untitled5.m 1.1
Workspace
Name Value Size Class
d 2 1×1 double
f [900,3500,28... 1×3 double
PL [97.5455,109.... 1×3 double
untitled * untitled2 untitled3 * untitled4 * untitled5 * untitled6 *
clc; clear; close all;
d = 2; % fixed distance (km)
f = [900 3500 28000]; % MHz
PL = 32.44 + 20*log10(f) + 20*log10(d);
figure
% -------- Graph 1: Frequency vs Loss --------
subplot(2,1,1)
plot(f,PL,'-o','LineWidth',2)
title('Propagation Loss vs Operating Frequency')
xlabel('Frequency (MHz)')
ylabel('Propagation Loss (dB)')
grid on
% -------- Graph 2: Band Comparison --------
subplot(2,1,2)
bar(PL)
title('Comparison of Propagation Loss Across Bands')
xlabel('Frequency Bands (Index: 1=Sub-1GHz, 2=Mid-band, 3=mmWave
ylabel('Propagation Loss (dB)')
xticklabels({'Sub-1GHz','Mid-band','mmWave'})
grid on
Figure 1
Propagation Loss vs Operating Frequency
Propagation Loss (dB)
Frequency (MHz)
×10^4
Comparison of Propagation Loss Across Bands
Sub-1GHz
Mid-band
mmWave
Frequency Bands (Index: 1=Sub-1GHz, 2=Mid-band, 3=mmWave)
Command Window
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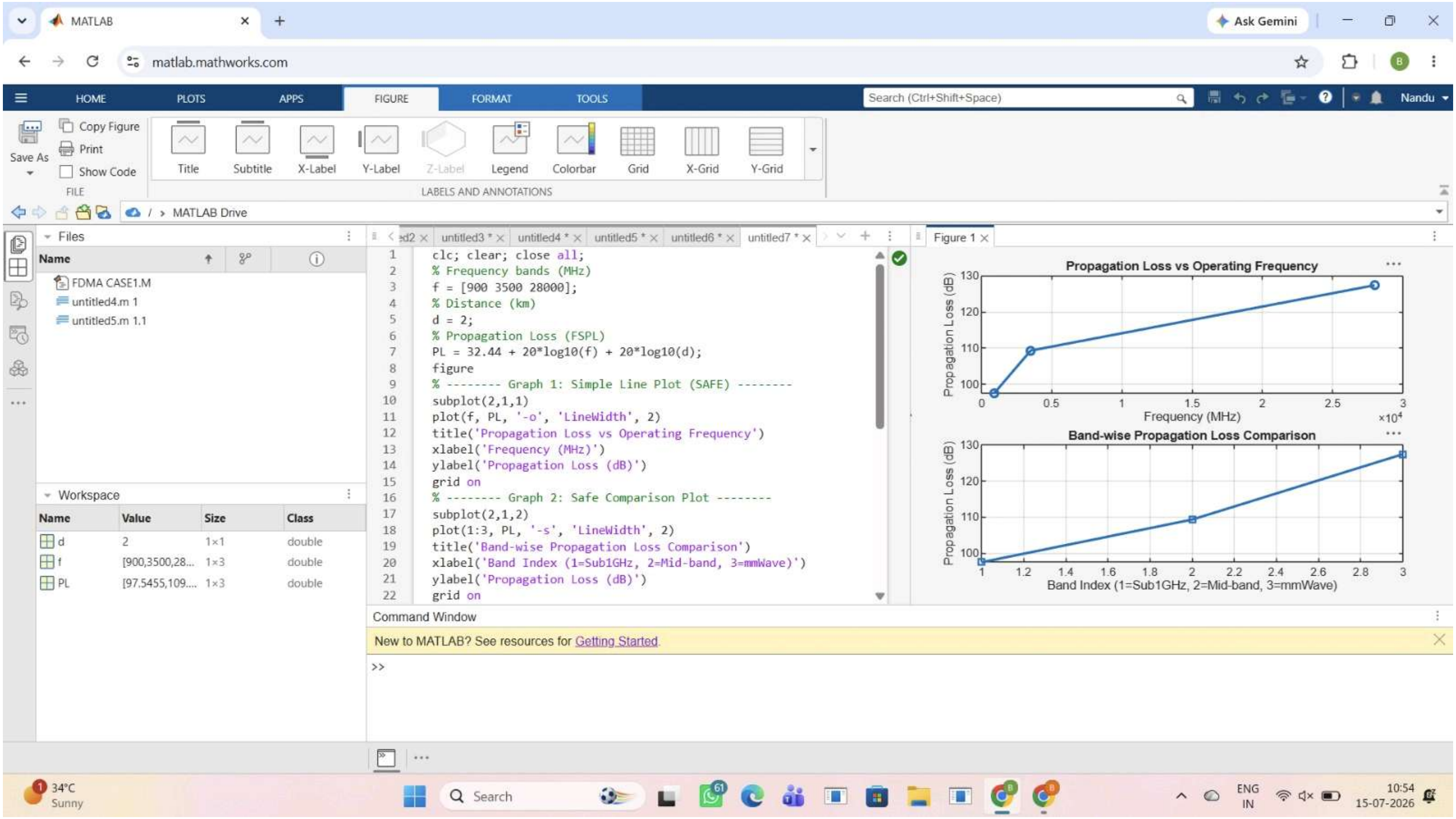
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untitled5.m 1.1
Workspace
Name Value Size Class
d 2 1×1 double
f [900,3500,28... 1×3 double
PL [97.5455,109.... 1×3 double
untitled3 * untitled4 * untitled5 * untitled6 * untitled7 *
clc; clear; close all;
% Frequency bands (MHz)
f = [900 3500 28000];
% Distance (km)
d = 2;
% Propagation Loss (FSPL)
PL = 32.44 + 20*log10(f) + 20*log10(d);
figure
% -------- Graph 1: Simple Line Plot (SAFE) --------
subplot(2,1,1)
plot(f, PL, '-o', 'LineWidth', 2)
title('Propagation Loss vs Operating Frequency')
xlabel('Frequency (MHz)')
ylabel('Propagation Loss (dB)')
grid on
% -------- Graph 2: Safe Comparison Plot --------
subplot(2,1,2)
plot(1:3, PL, '-s', 'LineWidth', 2)
title('Band-wise Propagation Loss Comparison')
xlabel('Band Index (1=Sub1GHz, 2=Mid-band, 3=mmWave)')
ylabel('Propagation Loss (dB)')
grid on
Figure 1
Propagation Loss vs Operating Frequency
Propagation Loss (dB)
Frequency (MHz)
×10^4
Band-wise Propagation Loss Comparison
Propagation Loss (dB)
Band Index (1=Sub1GHz, 2=Mid-band, 3=mmWave)
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